

Public Safety and Military Applications

Recent world events have heightened awareness of military and public safety functions in society. A great amount of resources have been applied toward ensuring the safety and security of our nation, its citizenry, natural resources, and treasured landmarks. Virtual reality is being applied in a variety of ways to contribute to the well-being of our nation. One of the highest uses of any technology is to save lives. Virtual reality is used to help save lives indirectly by training firefighters, police, and military personnel, as well as in educating the general public, in how to respond in various dangerous situations ranging from natural disasters such as tornados and earthquakes to acts of terrorism.

Virtual reality and related technologies such as telepresence contribute to saving lives directly by insulating safety personnel from danger. One example is using telepresence to allow military personnel to carry out dangerous missions from the safety of a remote location.

It is interesting to note that many applications related to public safety and the military have been manifested as entertainment applications. For example, many virtual reality games are of the “find the villain and shoot them” genre, and ideas from them can readily be adopted for more serious purposes. Likewise, VR hardware developed specifically for gaming can be adapted for use in public safety applications. One example is the *Total Recoil* system that was developed for the *Virtuality Trap Master* application (described in Chapter 9). The *Total Recoil* system includes a life-like shotgun that provides haptic feedback via a somewhat realistic “kick back” when the gun is fired.

Many applications appropriate for public safety and military have a requirement of being very closely coupled to the physical world. As such, haptic feedback and locomotion feedback devices can be very important. Thus, many public safety and military applications are coupled to real-world input and output devices such as treadmills, bicycles, and other devices to provide a realistic sense of physical exertion.

Due to the importance of learning to do tasks correctly, it is important for VR developers to deliberately avoid any problems with the VR system that could lead to ineffective training. This means carefully studying and implementing any simulations that are included in the application, eliminating harmful lags, and other artifacts of interacting in a virtual world rather than the real world.

7.1 AREAS OF APPLICATION

7.1.1 Equipment operation training

This area of application is similar to equipment training in other fields such as the Motorola Assembly Line Trainer. Much like commercial pilots learning to fly a Boeing 747 in a flight simulator, military pilots can learn how to fly an F-117A, and they must learn not only the flight control operations but also combat maneuvers under military conditions. An added element is the ability to practice specific missions such as a particular bombing mission. There is an element of training to learn to use a specific piece of equipment and another element in which the participant learns to do certain procedures with that equipment.

Situational training In many fields, split-second timing is crucial, so one doesn't have time to make plans and rehearse for the action (e.g., firefighting); in other situations, while the action might require split-second choices, there is time to prepare for the particular situation. So, for firefighting, it is beneficial to learn about how to handle many, unrehearsed situations, whereas for military or hostage rescue actions, one can spend time to prepare for a specific operation. Of course, in many cases, the same application could be used for both general training and for a specific operation (e.g., the *VRaptor* hostage rescue).

Mission planning A specific type of VR training that can be applied to any kind of route planning or maneuver planning can be referred to as *mission planning*. This can range from presurgery planning to the practice of a rescue maneuver. It could also include the repair of an expensive piece of equipment, such as the mission-planning VR application discussed in

this chapter in which astronauts planned for and practiced repairing the Hubble Space Telescope.

Why should a soldier endanger their life by being in a tank on the battlefield when through the use of telepresence they can perform the same operation from the safety of a remote location? Technology has been put to use in the military and in the local police department. The picture below shows a robot equipped with a camera, shotgun, microphone, speaker, and other facilities to assist in seeking out villains in hostage situations while the officer remains safely out of harm's way.

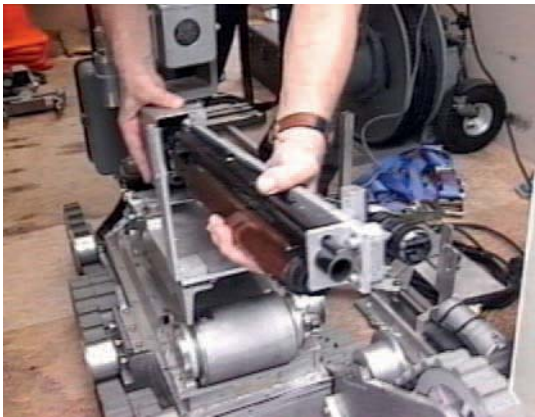


FIGURE 7-1

Police use remotely controlled robots in dangerous environments.

Human-guided bombs, with the operator located at a safe distance, are an example of a first-person point of view telepresence application. The operator is situated as though they were riding the bomb and guiding it down to the target.

Augmented reality has become more prevalent in military and public safety applications. As firefighters enter a burning building, they may have additional information displays that indicate where they are in the building floor plan, where volatile materials are stored, or where the home base indicates that victims are stranded in the building. The military has used heads-up displays for pilots and foot soldiers to provide additional information about their situation for quite some time. Night-vision goggles have been used to provide a view of the environment around the soldier that has been “augmented” by processing the image to allow them to see the area as though there is much more light present.

As technology progresses and applications are developed, virtual reality, telepresence, and augmented reality will continue to provide a vital role in helping to maintain the safety and security of the world. Whether indirectly via providing training or practice, or directly by keeping people out of harm’s way, virtual reality can play a role in making the world a better place.



FIGURE 7-2

Primitive methods of bomb guidance often required the operators to be in harm’s way. Virtual reality allows the military remote control from safe environments.

CASE STUDY 7.1: OFFICER OF THE DECK

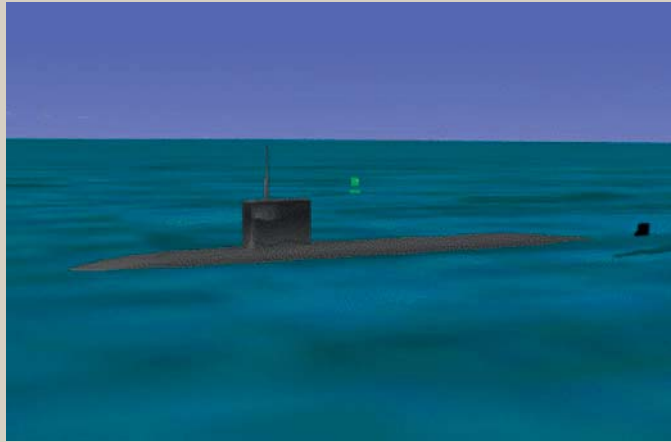
Executive Summary

The Officer of the Deck (OOD) application is a specific instance of a larger research effort exploring the use of virtual reality in training. The OOD application provides a simulation in which a naval officer can rehearse navigating a submarine safely into port. The task is interesting in several ways. One is that the task can seldom be rehearsed outside of the virtual environment. Additionally, the ramifications of a failure in the task can be disastrous in terms of safety and expense.

The OOD application mimics the way the task is carried out in actual practice. Primarily, the officer controls the

submarine via commands spoken to a simulated helmsman. The OOD application utilizes voice input to control the virtual submarine. The officer wears a head-mounted display and can see the harbor and navigational markers. The officer can also call for a variety of tools that emulate the real-life tools he would use.

An informal survey indicated that the OOD application shows that VR can be a useful tool for tasks of this nature. An interesting potential use would be to carry a portable version of the training system onboard an actual submarine and use it to rehearse harbor navigation while at sea. In fact, Advanced Marine Enterprises was recently

**FIGURE 7-3**

Bringing a submarine into harbor is a demanding task that requires nerves of steel and adequate training. A virtual reality simulator allows officers to practice the maneuver without risking billions of dollars.

awarded a grant to create a prototype *OOD* simulator based on the design work described here.

Introduction

The Virtual Environment Technology for Training Project (VETT) of the Sensory Communication Group at the Massachusetts Institute of Technology (MIT), and Bolt, Beranek and Newman Inc. (BBN) has investigated application areas for which VR training might be useful. Applications that have been developed include navigating a submarine to port, electronics troubleshooting, and others. Of these, we will look at one that has shown great potential for actual use in the military.

The *Officer of the Deck* VR application allows junior submarine officers to be trained in the experience of bringing a submarine into port. This is an especially useful task to train, because opportunities for learning this skill occur so infrequently. Submarines are brought to port only a few times per year, and each time, only one person has the chance to navigate the craft down the channel. In fact, only a few people can even be stationed outside the sub, on the conning tower.

The overall goal of VETT is to research methods of training using VR technology that will provide cost savings and/or improved learning. Anticipating that computer and VR

technology will continue to become more cost effective, and portable, the researchers on this project hope to find training tasks that demonstrate good ways of providing cost effective training via this technology. Some research areas of interest include investigating human adaptability to distortions, delays, and noise of typical VR systems; the relationship of VR technologies to motion sickness; and the use of haptic display as a human interface. In general, they have focused on training techniques, cues, interventions, and evaluation methods.

The specific goal of the *OOD* experience is to provide the opportunity for training a novice to perform a task. Considering that these novices are responsible for bringing a multibillion-dollar craft into port, eliminating human error is of the utmost importance. The rest of this section will focus on this pilot task of the VETT project.

Performing the role of officer of the deck (OOD), which includes the task of guiding a submarine to port, is a requirement for promotion of junior submarine officers. They must accomplish this task at least once, yet the opportunities to perform, or even witness, this operation are fairly rare. It is only done during scheduled stops to port. At the same time, errors can be extremely costly, causing hundreds of millions of dollars in damage, and jeopardizing the careers of all the boat's officers.

The actual task involves calling out piloting commands. The OOD is responsible for monitoring the submarine's progress along the marked channel, as well as observing other traffic and conditions. Currently, the training for this task is minimal.

Application Description

A specific scenario was implemented by the VETT project to test the use of VR as a training tool for this task. The task involved only the "in-harbor" portion of navigating a surfaced submarine into King's Bay port in Georgia, but did not include the actual docking operations. This real-world location is simulated in the computer-generated environment, as is a below-deck "piloting team."

This task was selected for a variety of reasons. It can be readily implemented in a fairly portable VR system. It can tolerate the constraints imposed by affordable VR systems such as lag and somewhat simple virtual worlds. It was a task for which most subjects would not have practiced, yet for which there were enough experts available to help validate the VR experience. Also, the task was of high interest to agencies providing funding.

Viewing the virtual world in a head-mounted display, the trainee monitors the navigation aids and progress of the boat, gives spoken commands to the (virtual) helmsman, and receives back verbal confirmations. Using this method, the trainee must successfully navigate the submarine through the harbor channel to port. In-harbor submarine movement is slow, and there is a natural lag in the real-life task between giving vocal commands and the visual response. This response can be matched by the simulated response, maintaining accuracy in the task interface.

Matching the lag is important, because task realism in support of the training is a key goal. Not only does the participant interact with the simulation in the same way they would in the actual task, but the space that the officer has to move freely in the real task is constrained, which gives justification to similar constraints imposed by the VR system. In an HMD VR system, it can be dangerous to give the user freedom to walk about a large area. In this task, a mock conning tower, constructed of metal pipe, acts to protect the immersed user, to keep them located under the



FIGURE 7-4

The King's Bay, Georgia, harbor provides the setting for the Officer of the Deck simulation.

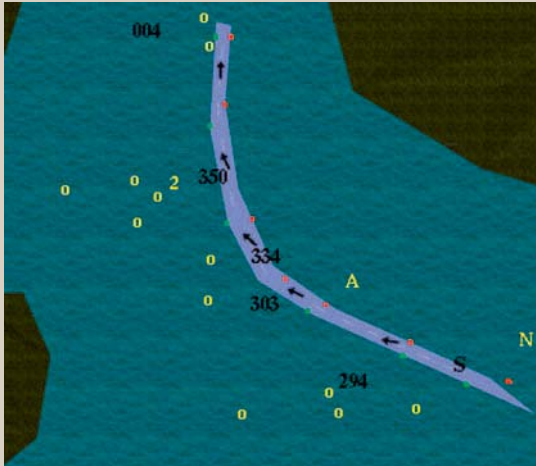
tracker transmitter, and to provide a prop that replicates part of the virtual world.

Representation of the Virtual World

The harbor of King's Bay, Georgia, was modeled for the initial experiments of the training system. The surrounding terrain is fairly flat, resulting in a model with relatively few polygons. The shoreline itself put the highest demand on polygon usage. Some of the shoreline contour features were reduced to keep the frame rate at the desired 15 frames per second, resulting in a database of about 5,000 triangles.

Important features of the bay were gathered from the Defense Mapping Agency (DMA) database and nautical charts. These features include the terrain, buoys, range markers, and turning beacons. No other maritime traffic moves through the simulated channel during the task. The experimental designers decided that movement of other traffic along with modeling the associated radio traffic was unnecessary to simulating the task, and would only complicate the experimental analysis.

Navigation through a current was one of the primary skills being trained. A current is modeled, causing the submarine to drift across the channel. This is visually represented by the wake of the buoys. Other than these currents, weather

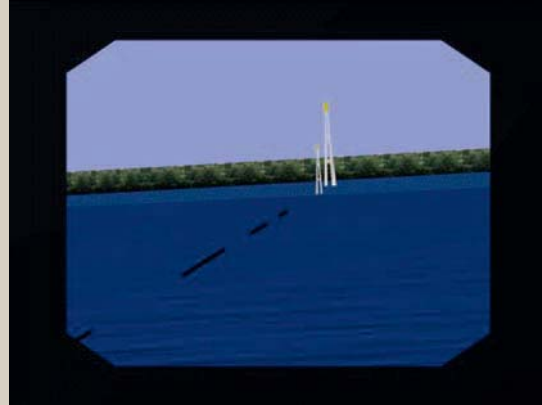
**FIGURE 7-5**

A “God’s-eye view” allows trainees to see the course they took through the channel during a postsession debriefing.

details did not need to be modeled. Submarines are only brought to port under good weather conditions when it is clear and with low wind. Again, the constraints of the real-life task make modeling the virtual world easier.

Tools that the trainee can use to help navigate through the channel are also made available as visual displays in the HMD. A compass display provides directional information in a virtual heads-up display (HUD). Also, the user is provided with a set of virtual binoculars that cause the user’s view to zoom in on distant objects (of course, just as with real binoculars, this magnifies head rotational movement, forcing the user to concentrate more on maintaining head stability). In one experimental condition, the current was also visualized (nonrealistically) as vector arrows on the surface of the water. (The preliminary experimental results did not show any significant benefits from using this display.)

In the real-world task, the OOD brings a laminated “course-card” to the conning tower containing a list of the channel segments. For each segment, the course-card gives the pertinent navigational information regarding which landmarks indicate travel down the center of the segment (range finders), and which landmarks indicate when to begin turns (beacons). In the virtual environment task,

**FIGURE 7-6**

Pairs of range markers are used to determine where the submarine is with respect to the centerline. In the VR display, the range markers are “perceptually tuned” so the trainee will be able to see them from the same distance as in the real world.

the spoken command “course card” invokes the display of a virtual course card in the HMD, replacing the display of the bay.

The primary goal of this experience is to design and understand the effectiveness of training in a VR environment. Realism is always subservient to this requirement. Details where realism was deemed important were rendered realistically; those that were not (such as trees) were not. In general, uses of photorealistic and audio-realistic representations of the world were used when possible as long as they didn’t conflict with the training effectiveness.

The visual scene that the trainee OOD sees is a low-resolution, cartoon-like polygonal representation of what truly can be seen from the conning tower of a submarine in the harbor of King’s Bay. The visual model includes the top of the submarine, the bay itself, important landmarks, navigational markers, and some optional visualization information such as flow vectors on the water and heads-up display information. Buoys are marked with identification numbers useful for verifying position. Range markers come in pairs of tall towers that indicate whether the submarine is in the center of the channel (or on which side) based on how they line up.



FIGURE 7-7

Buoys are used as a navigational aid. The number on the buoy can be cross-referenced to a chart that provides information on that buoy. Sounds are emitted by the buoys to help the officer determine their location.

The visual scene is displayed using the monoscopic-binocular method (two displays showing the same image). This requires less computational resources and does not detract much from the realism, since most of the objects in the scene (and all the important ones) are a great distance away, so the depth cue from stereopsis would not be very significant. Full 6 DOF position tracking of the head was implemented, so there are depth cues available from motion parallax.

Concern for creating a verifiable training application imposed the requirement for a method of overcoming the limitations of the display hardware.¹

Some objects in the scene might not “appear” realistic when they are a great distance away. *Perceptual cue tuning* is the general process of matching important virtual cues to natural perceptual cues. For example, training developers must make cues such as depth cues work the same way in the display as the participant experiences them in the real world, or compensate for the differences.

In this application, perceptual cue tuning is used to make distant objects become visible in the virtual world at approximately the same distance as they do in the real world. This is particularly important for landmarks and

man made navigational aids such as the buoys and range markers [Zeltzer and Pioch]. In the HMD display used in this application, these visual aids could not be “adequately identified” without making distant ones larger. This effect was dampened as the markers drew nearer, and not used once objects could be seen at “normal size” in the HMD. The application authors tested this method experimentally and found that it did indeed achieve the desired results.

DESIGN CHOICE

Visual Representation

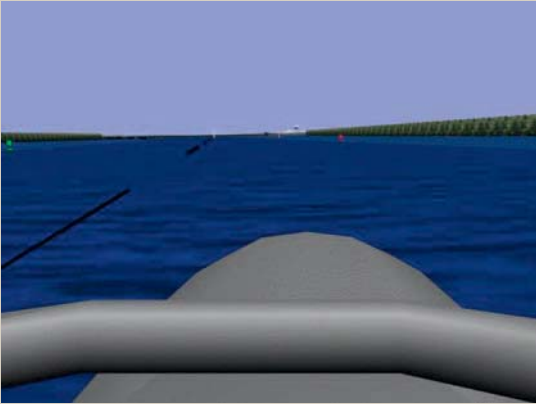
The application authors chose to use the technique of “perceptual cue tuning” to achieve a more realistic experience.

Audio cues were also provided to make the training session as realistic as could be feasibly implemented. The most important cue for navigation is the sound made by the buoys that mark the safe-passage channel. The two nearest buoys are selected, and 3D spatialized sounds are presented to the trainee indicating their relative location. Also important is the orders acknowledgement provided by the virtual helmsman who echoes the commands given by the OOD. Wind sound is added to the experience, providing an ambient cue that helps increase the sense of realism.

A “haptic display” is provided by the metal cage that represents the cage atop the conning tower. Though simple, this prop is effective at grounding the user in the virtual world and gives important nonvisual cues such as the direction of the bow of the boat (indicated by a knob placed in the center).

Vestibular and other sensory cues are not provided by this experience. Additions of these might increase the degree of immersion slightly, but not while maintaining the cost efficiency and portability of the current implementation. For example, one might consider the added benefits of a vestibular display of the rocking of the submarine, or of an ambient olfactory display of harbor air. However, submarines don’t rock very much when going to port, and the smell won’t vary much from the sea air the officers are used to experiencing.

¹Such as poor resolution.

**FIGURE 7-8**

The handheld rail in the VR simulator is registered with a pipe surrounding the participant providing both haptic feedback as well as a constraint to the spatial boundaries of the platform.

Interaction with the Virtual World

The *Officer of the Deck* experience is primarily a training application. Part of the requirements of this *genre* of applications is that it teaches a fixed set of skills such that the trainee is able to perform to a certain standard. The *form* in which this activity takes place is a pilot-through controlled via voice commands.

The *narrative* is generally typical of a training application. The trainee is given some pre-experience instruction, including a videotape. They then don the HMD, and begin the experience. To help the trainees achieve a sense of personal context in the virtual world, they are taken on a short predetermined (ride-along) path from outside the submarine to the top of the conning tower.

DESIGN CHOICE

In response to feedback from test subjects, the application authors chose to prepend a short, predetermined ride-along flight to the training experience to give more visual context to the trainees. Beginning the experience while the participant was standing on the conning tower was often confusing.

Debriefing of the subjects is the post-trial portion of the overall experience. During the debriefing, the subjects can be shown the path of their submarine during the last trial. This God's-eye view of their efforts in the virtual world constitutes another level of narrative. After "unsuiting," they are shown a 2D map of the bay on a video monitor. A red line indicates the path traveled during the exercise. This process allows trainees to evaluate their own efforts and get a better feeling for how to modify their commands to improve their performance.

Speech was used to provide a simple navigation interface in the application. A single handheld push-to-talk button is used to indicate when commands are being directed to the virtual helmsman. There are no other handheld controllers, only a single button to push when they wish to issue commands. This control mimics the real world, where the OOD must press a similar button to speak to the helmsman. The constraint of the cage even limits physical movement to small movements and turning the head. The rudder was visually represented, and the subjects could turn their heads to visually verify the rudder's orientation, though head motion was not measured in the task analysis.

Realism of the travel interface is important for training this task. In this case, that interface is through the verbal communication with the helmsman. Thus the virtual helmsman will respond only to properly phrased commands, and with the appropriate response. An example exchange might go something like: for the command "Left standard rudder," the system would respond with, "Left standard rudder, Bridge, Helm, Aye." Though the experiment designers assume the candidates have already been trained in basic ship handling, the appropriate lag for turn and throttle is simulated and can be witnessed visually.

There are no object interactions per se in the *Officer of the Deck* experience, but there are some visual tools that can be summoned. Like the travel interface, requests for a tool are also made vocally. Tools that can be summoned include the compass, binoculars, and course-card displays.

There are no complicated visual agents populating this virtual world. The only entity that could be considered an agent is the virtual helmsman, who can only be heard and who responds with explicitly defined responses based on naval procedures.

There are no collaborative interactions in the truest VR sense. This is a single-user training system. The only thing that might fall into this category is the actual experimenter, who controls the parameters of each trial, and might be able to control some aspects of the world during a trial. Were the helmsman responses generated by a real person, such an interaction might be considered multiple-user collaboration, but in this case, it would be an example of a surrogate agent.

The simulated physics of the virtual world were designed to be as realistic as needed for effective training, and due to the fact that the world is somewhat simple, good realism was achieved. A simple submarine propulsion model was implemented. The actual propulsion system is classified, so the model was estimated. Naval officers who had served on submarines confirmed that the implemented model was near enough to be acceptable for training.

The other world physics that might have required modeling are the weather and water conditions. Since submarines only go to port in good weather, no special modeling of weather and water conditions was required. Also, water current is not rapid, and is fairly constant in the middle of the harbor where the submarine stays unless the trainee is failing the task. The submarine remains at the surface of the water during the entire procedure, so underwater currents and physics do not need to be simulated.

Venue

The venue for the experimental analysis of the application was a laboratory facility of the Sensory Communications Group of the MIT Research Laboratory of Electronics. Actual training, however, can take place either at a Naval training facility or onboard the submarine itself. Each trial training session lasted about 20 minutes.

VR System

Two prime requirements of the OOD training system were that it be cost-effective and portable, perhaps portable enough to be placed aboard a submarine. Most of the system choices reflect these constraints.

SYSTEM CHOICE

Display Paradigm

An HMD was chosen as the visual display paradigm (the virtual research model VR4). The full field-of-regard of an HMD gives the user the ability to look from the submarine in any direction, and is a much more portable visual system than projection VR or even *BOOM™* displays. A full field-of-regard was a necessity because the OOD is constantly looking in all directions around the submarine as it navigates the channel. The requirements for portability and simulating the task to include the same viewing capabilities as available in the real world dictated that an HMD display be used.

The specific HMD model used was a moderately low-resolution model. Because of the low resolution of the display, the use of the perceptual cue tuning was required in order to see objects that take very little of the screen space. A Polhemus FasTrak magnetic system provides 6 DOF position tracking information of the head.

The world is rendered as monoscopic images at a frame rate of 15Hz on a 2-processor Silicon Graphics graphics Onyx™ workstation with the RE2 graphics hardware.

SYSTEM CHOICE

Visual Rendering

Because most of the virtual world is in the distance (everything but the submarine itself), stereoscopic images do not provide a considerable amount of depth information. Thus, it was not considered essential to render stereo images. Rather, monoscopic images were rendered instead.

SYSTEM CHOICE

Visual Rendering

The pace of the task is not very rapid. Therefore, extremely high frame rates are not required. The main constraint on frame rate is the lag between head motion and the appropriate view. A rate of 15 frames per second is high enough to make the tracking system the primary cause of lag. This allows more detail to be rendered than if a higher frame rate was required.

Audio input and output is handled via a small boom microphone attached to the front, and headphones installed inside the HMD. A handheld push-to-talk switch allows the participant to indicate when they are giving commands to the virtual piloting team, mimicking the real interface in which the officer also uses a push-to-talk switch to give instructions to the helmsman. The BBN *HARK*[™] speech-recognition package provides speaker independent vocal input to the application. The 3D spatialization effect is produced by the CRE *Beachtron*[™] system. A midi switching device is used to select which sounds are spatialized and which sounds are presented as ambient sounds in the environment. All the audio input and output generation for the speech communication are done on a Silicon Graphics Indy[™]. Other sounds were stored and played from an Apple Macintosh[™].

SYSTEM CHOICE

Audio Rendering

It is important to present some of the sounds in the virtual world spatialized in three dimensions. In particular, it was important to spatialize the sounds that come from the navigational buoys that help trainees to locate themselves in the world.

Application Implementation

This application was developed as a specific product for use by the United States Navy. As such, a complete version

was finalized (version 1.0). One specific feature that was not included in version 1.0 but which the developers felt was important was to add a collection of artificial instructional cues. That is, information provided by an agent in the world who can aid the trainee when they require it.

Development on version 1.0 ran for about 1 year. It began in February 1994, when the VETT project was first asked to create the application. By August of that year, a prototype version of the experience was running, and evaluated by Naval officers with submarine experience. Version 1.0 was completed in February 1995.

Many people were involved in the creation of the *Officer of the Deck* virtual environment, including principal investigators David Zeltzer and Nathaniel I. Durlach. Important programming contributions were made by post-doc Jeng-Feng Lee; graduate students Walter Aviles, Nicholas Pioch, Rakesh Gupta, and Jonathan Pfautz; undergrad students Erik Nygren and Brett Reid, and staff programmer Dorrie Hall. The *OOD* simulation was developed with the close collaboration of the BBN Training Team that included Bill Levison, Yvette Tenney, Dave Getty, and Dick Pew. Work on the *OOD* was sponsored in part by the Naval Air Warfare Center Training Systems Division and the Office of Naval Research.

The VR experiment for the *OOD* training application was put together by a team of researchers from MIT and BBN under the direction of principal investigator David Zeltzer. The VR system and *OOD* application were developed at MIT. BBN helped with the design of the experimental task and provided the speech-recognition tool.

There are many software subcomponents for a VR training application like the *Officer of the Deck*. In addition to the graphics and audio rendering, and the simulation dynamics, there were components to handle speech recognition that provided the world modeling interface, and the experimenter/instructor interface. The experimenter interface was the mechanism for controlling the experimental and training parameters, as well as recording the trial data.

The graphics software components were written in C using the *Performer* graphics library from Silicon Graphics. The submarine and world dynamics simulations were written in C++.

Related Applications and Follow-on Work

Officer of the Deck training continues to be an important part of naval training. One of the most interesting related VR-based training systems is the *Virtual Environment for Submarine OOD Ship Handling (VESUB)* project originally deployed at Submarine Training Facility in Norfolk, VA, and the Naval Submarine School in Groton, CT. Now developed by RDR, Inc. of Centreville, VA, the *VESUB* trainers are deployed at six major submarine training facilities throughout the world.

The *VESUB* trainer utilizes a HMD and provides voice-synthesized feedback in addition to speech recognition. The system has been through thorough assessment, and through this feedback it has been continually improved (see Vincenzi 2003, Hays 2000). Hays and Vincenzi found that over 90% of observers and participants felt that the system provided an increase in ship-handling skills.

Conclusion

The *Officer of the Deck* application was tested using experiments designed by the BBN training team. Subjects for the pilot experiment were MIT students who generally have a similar aptitude as Navy officers, though the MIT student subjects had little or no previous boating experience. For the final experiment, naval submarine officers were used, though these were not officers whose position would call for them to be an OOD.

In the experiment, each subject was given limited background instruction and system familiarization, after which they performed 10 trials on the training system. Measured performance improved significantly over the 10 trials. Two measures that indicated the improvement were the time spent outside the specified channel (average of 3.5 seconds in trial 1 to zero in trial 10), and the root-mean-square (RMS) distance from the channel centerline (from 28.5 yards to 6.9 yards). These results appear to validate the VR experience as a training tool.

Unfortunately, it is difficult (nearly impossible) to compare the results with a control group. This difficulty is because submarines are hard to come by and there is a general reluctance by naval officers to be measured in task performance.²

²Because of the possibility of its use in determining promotions.



FIGURE 7-9

The VESUB application continues to train submariners at several military sites.

Fortunately however, the *VETT* project team had a close working relationship with some U.S. Navy officers who were familiar with the task. Some of these officers participated in controlled evaluation sessions, and all felt that this system had great potential as a training and mission rehearsal tool. These kudos by domain experts further validated the *OOD* experience for training, including the use of perceptual cue tuning to accommodate for the poor resolution of the HMD.

Since mishaps involving submarines coming into port are rare, it is hard to derive specific numbers of how much can be accomplished with this application. However, if it helps to prevent even one accident, then the number of lives (and careers) saved as well as the amount of money (hundreds of millions of dollars) could be significant.

As an experiment to demonstrate the potential for use as a training tool, the project was successful. This tool provides an opportunity for OOD candidates to experience pre-training on a task that otherwise provides little opportunity for apprenticeship. Task-learning of the (simulated) task was accomplished by experimental subjects. In addition, naval officers associated with the project feel that such a system aboard a submarine would be good for performing mission rehearsal prior to coming into a new port.

This experience is unlike many typical adrenaline-rush games. It involves a much slower-paced task and is not meant to be particularly fun, so it might follow that it wouldn't be very mentally immersive either. During the times when it was "smooth sailing" for the subjects, they probably were more aware of the surrounding (non-VR) environment. However, the experiment instructor (Dorrie Hall) felt that "when the situations demanded focus, they were *there*." The purpose of the training application is to involve the subject enough so that they actually learn something.

Some interesting observations made by the application design team include noting features of good training as

well as features of the VR system that may not be necessary. Regarding training, it was observed that features that make good performance-analysis tools do not necessarily make good training cues. In fact, they can become a crutch that ends up causing a negative correlation to task learning. Part of the VR system that might not be necessary for this and similar tasks (where all the action happens off in the distance) are comprehensive depth cues. The implementation for the experimental procedure did not include a stereoscopic display, but it did provide motion parallax. Not all depth cues are important for all tasks, but some cues are very important. Finding out what cues are important is vital for any training experience. The use of perceptual cue tuning was a significant part of this process. One (monocular) depth cue that they did find to be important was the perspective foreshortening of objects on the shoreline receding into the distance. The *Officer of the Deck* VR training experience demonstrated a good use of the medium to produce a cost-effective tool for a task that has traditionally had limited opportunity for training. The OOD development team put significant effort toward the validation and verification of the experimental task, following the advice of Fred Brooks: "Build the right product, and build the product right."

Perhaps a key validation of the overall experience is that all the officers of the U.S. Navy who they had worked with during the project "expressed enthusiasm about the simulator's potential usefulness for training and mission rehearsal." [Zeltzer et al.]

CASE STUDY 7.2: SHADWELL VR EXPERIENCE

Firefighting Trainer

Executive Summary

Not surprisingly, the Navy is interested in utilizing optimal firefighting procedures for shipboard firefighting. The Naval Research Laboratory (NRL) maintains a decommissioned ship for use as a research vessel for exploring improved firefighting techniques and training, the ex-*USS Shadwell*. One of the NRL's research projects has been

to explore the potential effectiveness of VR as a tool for training shipboard firefighting techniques, mission planning, and rehearsal. The Advanced Information Technology and Navy Technology Center for Safety and Survivability branches of the NRL teamed up to create an experiment to study this issue.

Many different experiments on firefighting techniques are conducted aboard the *Shadwell*. The instructors on the

Shadwell have the ability to create controlled fires that simulate ones that firefighters are most likely to encounter. The use of VR for mission rehearsal would allow firefighters to familiarize themselves with a new ship while attempting various firefighting techniques without endangering people or ships.

Because the *Shadwell* was already being utilized for the study and testing of firefighting techniques it was also the site chosen to conduct feasibility tests of using VR as a supplement to the mission rehearsal process. A battery of such tests was held aboard the *USS Shadwell* September 18–22, 1995.

Application Description

To create the virtual training environment, portions of the *Shadwell* have been modeled at full scale in the VR environment. The application developers created a realistic model of the ship utilizing photographic texture maps to provide surface gradients for the bulkheads and decks. Objects that could be used in the firefighting procedure were modeled as 3D virtual objects. Other objects, with which the trainees would not be required to interact (such as the oxygen-breathing apparatus racks), are each represented as a simple textured polygon. A handheld input device used for interacting with the virtual world is represented as an avatar of the user's hand. No aural representation of the world was created for the initial experiment.

Two significant challenges in the fighting of fires are lack of visibility due to smoke and lack of familiarity with the area where the fire is located. Thus, a reasonable representation of smoke and fire are required to enable the participants to practice under these conditions. The *Shadwell* VR experience thus contains a simulated fire, complete with smoke plume and ambient smoke. The simulation controls the growth of the fire, as well as the smoke plume emanating from it. The fire-and-plume display was implemented as a loop of texture maps on a viewer-facing polygon. The textures were created using video of an actual fire. Ambient smoke was created using the atmospheric effects of the Silicon Graphics *Reality Engine 2*.

Collision detection was employed so users would be forced to move through the ship in a manner similar to real life. Thus, the firefighters would need to follow the same



FIGURE 7-10

Computer-generated smoke gives firefighters realistic clues regarding the location of a fire, as well as obstructs their view.

path in the virtual world as required by the real world constraints, including pausing to open doors, etc.

To open a door, a procedure was implemented that substitutes for the real-world task of manipulating the latches, wheels, levers, etc., normally required for opening ship doors. In the *Shadwell* VR experience the participant points their hand avatar at the door and presses the appropriate button on the device to open or close the door. The door stops moving when the button is released; thus the participant can control how far the door is opened or closed.

The primary travel metaphor is a *walkthrough* using a pointer-directed *walkthrough* interface. Use of this metaphor allows participants to look around while traversing the ship. This is an important feature, because they can make note of important objects and landmarks while moving through the *Shadwell*. The pointer-directed interface replaced a previous model that used a gaze-directed *walkthrough* interface. In the gaze-directed interface, the inability to look around while walking longitudinally proved to be ineffective. The *walkthrough* interface is important since the goal is for the participants to learn to travel a specific path through the ship that they will later have to traverse on foot. A terrain following algorithm moves participants vertically as they traverse stairs and ladders.

**FIGURE 7-11**

As in the real world, firefighters must pause to open doors aboard the virtual USS Shadwell.

**FIGURE 7-12**

Terrain following allows firefighters to traverse stairs and ladders in the virtual world.

The model of the world used in the experimental trials replicated portions of the *Shadwell*. A similar world was also created as a test model resembling the ship, but not representing any actual layout. This allows participants

to become familiar with the virtual reality user interface (before performing the actual experimental trials) without allowing them to become familiar with the ship in the test environment.

To evaluate VR as a tool for preparing firefighters and performing mission rehearsal, a feasibility test was implemented with two experimental groups. Subjects for the test were chosen from the expected group of users, trained Navy firefighters. None of the firefighters had any previous experience with the *Shadwell*. Twelve subjects were selected and divided into two groups, with equal gender representation between groups (four men, two women). The control and experimental groups were both prepared in the traditional method, including the study of ship layout diagrams. The experimental group was given additional preparation in the VR environment.

Each VR experience was presented to the VE training group in three stages. The first stage was to tour the ship in a ride-along travel method. This was to allow the participant to locate landmarks and useful objects on the course without needing to worry about navigating. As they traveled, a narrator (the instructor) drew attention to the important features along the path. Next, the participant used the point-to-move style *walkthrough* travel interface, and manually navigated to the specified location. In the third stage, they again had to navigate their way through the course, but now with ambient smoke limiting their visibility.

Both groups were tested in two different tasks. The first task (phase I) was a navigation-only task involving no firefighting skills. The second task (phase II) involved a complete attack on controlled fire, including locating the equipment, preparing the attack team, and extinguishing the fire. After standard mission planning, and VR rehearsal for the experimental group, both groups were tested on the actual *Shadwell*. For each test, they were measured on their ability to accomplish the navigation task (in phase I) or firefighting task (in phase II) on the actual *Shadwell*. The primary measurement was the time required to accomplish the task. A count of wrong turns was also recorded.

The results showed measurable improvements in the group with VR rehearsal versus the non-VR group. Though not statistically significant due to the small number of subjects, the preliminary results indicated that the VR prepared group was able to complete each task in substantially less time, and with fewer wrong turns than the control group. Details of the study were described by Tate, Sibert, and King in "Virtual Environments for Shipboard Firefighting Training" [Tate et al.].

After the testing, the VR participants commented that the VR component of the training was very helpful and that the fire was quite realistic. Those who did not receive the VR component felt unprepared when they entered the unfamiliar ship.

Overall, this experiment indicates that VR has potential as a tool for shipboard firefighting preparation and could be used to provide a number of training scenarios. However, studies involving more subjects are needed to give statistical significance.

Areas the experimenters feel need further research and improvement include interaction techniques for manipulating objects (as well as studies for validation of those techniques) and the integration of other I/O such as 3D sound, tactile (in the form of heat display), speech input, integrated multimedia and hypermedia instruction, and multiuser interaction.

VR System

This feasibility test was carried out directly on the *Shadwell*. The application was originally designed for the subjects to be standing during the experience. However, to accommodate the tracking system in a room with a low metal ceiling, the transmitter had to be lowered by 2 feet. So, subjects were seated on a stool. The stool could swivel, allowing participants to face any direction.

This project was led by David Tate of the NRL, who also did the user interface design and model construction. Tony King provided the software design and implementation,

and Linda Sibert performed the experimental design and analysis. The NRL virtual reality lab is directed by Larry Rosenblum.

Development for this specific application took approximately five months, though it was based on work brought to the project by Tony King and Perry McDowell. Graphics programming was done using both the Silicon Graphics Performer 1.2 and IrisGL libraries.

The world is presented to the participants using a Virtual Research VR2 head-mounted display. Tracking of the HMD and handheld unit is done with the Polhemus FasTrak system. The Naval Research Lab also has a Responsive Workbench, and has installed a CAVE-like projection VR display. However, an HMD was used for this application because it works well with this application both because of the restrictions imposed by the venue, and the desire for a full field-of-regard display.

Conclusion

The *Shadwell* VR training experience developers have begun work with the Naval Surface Warfare Center in Philadelphia on the development of a simulated shipboard firefighting system. Some enhancements for future versions of the experience include the use of props as input devices that provide a better corollary to the real-world controls, and the addition of 3D sound. The input control will be a virtual hose nozzle with adjustable spray pattern and water control modeled to mimic the real world, allowing users to realistically control the virtual water, with an enhanced simulation of water in the virtual world. Other enhancements in this version will be the addition of 3D spatialized sounds of the fire, water, doors and footsteps. Their expectation is that this system "could be used not only as a training device in classrooms, but also for shipboard refresher training or rehearsal."

The application developers "foresee using VR as a standalone training tool where firefighting procedures can be taught through simulation of fire growth and extinguishment."

CASE STUDY 7.3: SANDIA'S VRAPTOR

Hostage Rescue

The Virtual Reality/Intelligent Simulation (VR/IS) team at Sandia National Laboratories created a system for *VR Assault Planning Training or Rehearsal (VRaptor)*. *VRaptor* is a system that provides the capability for end user instructors to create scenarios for situational training, including options to manipulate the environment as the training is taking place.

The VR/IS team was primarily interested in studying the scripting and control of virtual actors. The *VRaptor* system is interesting in that it demonstrates that groups such as security and emergency response forces will be able to practice situations for which they might not have a chance to rehearse in any other way.

Application Description

For testing the usefulness of the *VRaptor* system, a hostage-rescue mission was created as a demonstration scenario. Specific training scenarios, such as a “room clearing operation,” typically consist of a set of defined steps such as breach the room, throw in a stun grenade, enter the room, command the room occupants to “get down,” and shoot the armed adversaries. One of the crucial decisions

for the trainee to make is whether to shoot or not and at whom. The trainees must avoid shooting the hostages and getting shot themselves. The instructor can subject the trainee to a similar scenario several times with the virtual actors in different roles. Another use of the system is to rehearse an actual task to find an optimal strategy before completing the action in a live situation. The real-world method of preparing for these tasks (either as general training, or for a specific operation) is through the use of a “shoot house.” A “shoot house” is a room modeled to represent the location of interest. The shoot house room is populated with “cartoon drawings or mannequins” representing the individuals involved in the scenario. Because the drawings and mannequins can only move in very simple ways, the opportunity for creating interesting situations is limited.

In the VR training paradigm, the instructor first creates the environment, places the virtual actors in their starting positions, and assigns scripts to the virtual actors to control their behavior. The trainee follows the assigned procedure while the virtual actors play their roles and realistic sound effects are triggered. When the scenario takes place, the action is sometimes very fast.

The virtual world of the hostage scenario consists of a room with furnishings, the avatar of the trainee, and virtual actors represented as human forms. The instructor can script the reactive behavior of the virtual actors. They can also trigger high-level behaviors for the actors while the scenario unfolds.

Sound effects help increase the realism of the event and augment the mental immersion felt by the participant. In the demonstration application described here, the sounds of the stun grenade and gunshots are the only ones heard.

VR System

The application interface consists of an HMD display for the trainee, and a view window and control panels for the instructor. The participant holds a replica of a 9mm Beretta, which is tracked for 6-DOF position as well as trigger pulls and when the participant inserts or removes the clip. The gun is realistic in feel and weight, but does



FIGURE 7-13

The VRaptor system provides a relatively realistic environment for hostage rescue training.

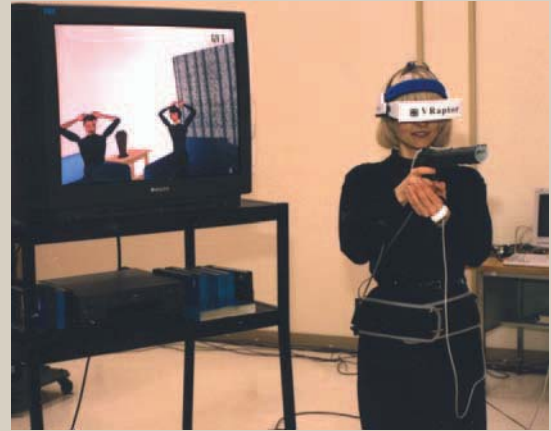
(Image courtesy of Sandia National Laboratories)

**FIGURE 7-14**

A trainee must make quick decisions as to whether or not to shoot. Image courtesy of Sandia National Laboratories

not recoil when fired. A position tracker is mounted on the gun replica and another unit is affixed to the HMD for head tracking. The participant's other hand is also tracked, as well as their lower back. These additional sensors allow both hands to be represented in the participant's view and a more complete representation of the posture of the participant in the external 3D view on the trainer console. Multiple third-person views and trainees are also possible, but were not incorporated into the demonstration system.

The instructor interface is on a multiwindow graphics workstation. One window is a three-dimensional third-person view of the virtual world including the avatar of the trainee. In another window, a set of menus allows the instructor to do activities such as assign the roles of the virtual actors (agents) to be hostages or terrorists in the next scenario. The instructor places the actors in the scene and selects from a list of scripts that describe the behavior an actor will take in different situations. One selection is what a virtual actor will do when shot at. The instructor can perform "Wizard of Oz" agent control by initiating high-level behaviors via menu interaction as the action plays out. The menus are context sensitive, providing only choices that make sense according to how the actors are positioned.

**FIGURE 7-15**

The trainee holds the weapon in front of herself. The monitor reflects the image that is being viewed by the trainee in the HMD. Image courtesy of Sandia National Laboratories

VRaptor is a system in which environments can be modeled and populated with virtual actors who respond according to the actions of the trainees. Although the developers do not see the *VRaptor* system as replacing the physical shoot house entirely, they do believe this VR version of a shoot house offers an option that can augment this live training, providing additional features, such as virtual characters that behave according to programmed actions. It also offers a lower-cost option to using a physical shoot house.

The specific work for the *VRaptor* demonstration application took approximately 4 months. However, it was built on the VR software platform under development at Sandia National Laboratories. The Sandia VR software was originally written in *IrisGL* and later ported to Silicon Graphics' *Performer* library.

The VR/IS team also integrated elements developed at other computer graphics labs. These were the geometry from the *Jack™* system, the *KPL* language from Ken Perlin's research on behaviors for virtual actors, and *Simlet Designer™* from Inflorescence Inc. A client/server audio system developed at Sandia was used to combine sounds into a single output.

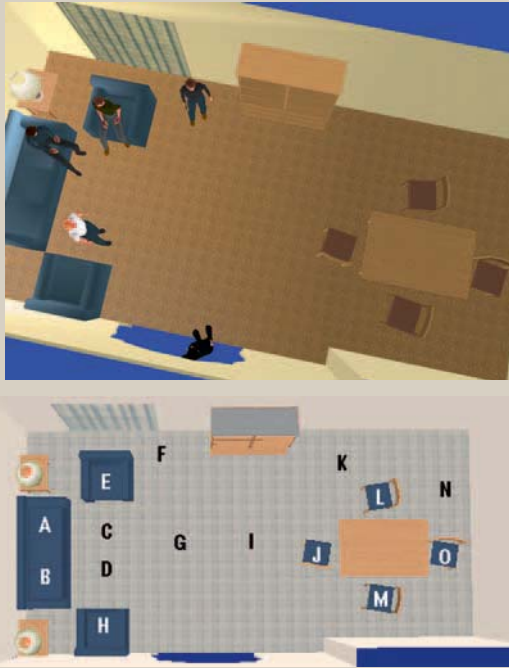


FIGURE 7-16

The instructor interface allows actors to be positioned at any of 15 different locations. Whether an actor is a hostage taker or hostage cannot be determined by physical appearance. Their actions typically reveal their status.

Image courtesy of Sandia National Laboratories

The *VRaptor* project was conducted by Sharon Stansfield and Dan Shawver of Sandia National Laboratories. Jim McGee and Wade Ishimoto provided subject matter expertise in the hostage rescue operation. Denise Carlson, James Singer, Ron Hightower, and Monica Prasaad contributed to the development of various components of the technology.

An Optics One PT01 HMD with 640×480 resolution and a 40-degree FOV was used to display the world to the participant. A Polhemus electromagnetic-style position tracker with four sensors tracked the participants' movements. The graphics were rendered on a Silicon Graphics

workstation. Early versions ran on the Crimson™ model deskside workstations. Later versions have been run on an Onyx™ with Reality Engine 2™ graphics hardware and on a Silicon Graphics Octane™. The VR/IS team is in the process of porting the graphics software to a Microsoft Windows™ platform. Audio is generated on a Silicon Graphics Indigo2™ workstation.

Conclusion

In the future, the *VRaptor* developers would like to do an evaluation on the effectiveness of this system for training, but no formal analysis has yet been done. However the Sandia team does believe their system will be suitable for many operations. To achieve a system that would meet the needs of law enforcement officers, they worked closely with a trainer for the FBI's hostage rescue team from the onset of the project. They have subsequently demonstrated the system to personnel from many other law enforcement agencies, including the Los Angeles Sheriff's Department, Department of Energy security, and the military. The developers found that most of them stated that the system had the potential to be a valuable tool.

Participants who have tried the simulator have reported favorable results. One of the features the developers would like to improve is to raise the resolution of the control of the actors' behaviors. They also would like to monitor and log the trainee's behavior to allow the performance to be reviewed.

One of the significant findings of this work by the developers is that photorealistic graphics are not necessarily the most important aspect of creating a believable virtual world. Rather, a world that behaves and responds appropriately can provide a strong sense of immersion and suspension of disbelief.

In regards to how Stansfield feels VR will be useful as a training tool for law enforcement procedures, she states:

While virtual reality supporting such highly behavioral, interactive simulations is still in its infancy, we believe that its application to training in high-consequence environments such as this will prove to be one of the most important and useful.

CASE STUDY 7.4: INTEGRATED EVA/RMS VIRTUAL REALITY SIMULATION

Space Walk Trainer

The Hubble Space Telescope (HST) was launched in April 1990 to enable astronomers to gain a view of the heavens from a location clear of the distortions caused by the earth's atmosphere. Unfortunately, shortly after the deployment of the HST, the optical system was found to have disabling flaws. Repair of the telescope would require a manned space mission with multiple extravehicular activities (EVAs), that is, space walks. From 1990 to 1993 the National Aeronautic and Space Administration (NASA) put considerable effort toward the planning and crew training for a Space Shuttle mission to repair the HST.

The Simulation and Graphic Development Branch at NASA's Johnson Space Center (JSC) had been exploring VR applications in 1990 and was interested in using VR as a mission rehearsal tool. The VR training system they developed was a very ambitious project, and resulted in the first VR system used for preparing for an actual task in space.

One of the reasons they embarked on this project was to demonstrate the potential of VR as a mission-preparation

tool. The goal was to help the astronauts understand the situations that might be encountered while repairing the Hubble Space Telescope and to help develop the procedures to be used in the mission. Since this mission would involve more EVA operations than previous shuttle flights (five nearly 8-hour activities), it was important that the astronauts be prepared for all the different circumstances that might arise during the operations, and be able to avoid, or quickly avert difficult situations. It also provided an opportunity to develop a communications protocol between the two EVA astronauts and the Intravehicular Activity (IVA) astronaut who controlled the Remote Manipulator System (RMS).

In December 1993 (3 years after the launch of the HST), the telescope was successfully repaired on shuttle mission STS-61. The VR lab at JSC has continued to be used as part of the shuttle mission training program. Since this first mission, all subsequent EVA procedures have been simulated using the VR system, and the JSC "Integrated EVA/RMS Virtual Reality Simulation Facility" was born, becoming a permanent facility at the space center. A large number of shuttle missions with EVAs are being planned in preparation for the space station NASA is building in conjunction with many other space agencies.



FIGURE 7-17

Astronauts Jeffrey Hoffman and Story Musgrave on a space walk to fix the Hubble Space Telescope during shuttle mission STS-61.

Application Description

NASA has relied on simulators and specialized training facilities since the days of the Apollo program. This has worked well, but these unique facilities with large support staff are very expensive (generally exceeding \$30,000 per hour) and offer limited training scenarios, with limited availability. In addition to this, they cannot replicate the physics of microgravity well enough for all tasks. For instance, moving large objects in a water-tank is much different than in space, as a lighter touch is required in space due to the lack of resistance.

The only true way to simulate microgravity without leaving the earth's atmosphere is in the KC-135 aircraft (also known as the "Vomit Comet"). Even this is limiting because microgravity can only be experienced in 25-second durations, whereas in virtual reality an astronaut can be immersed in a simulated microgravity world for hours.

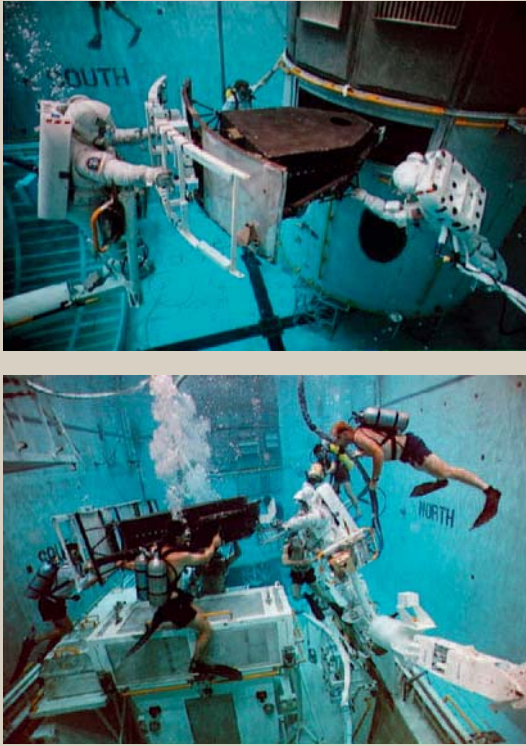


FIGURE 7-18

In the photo on top, Astronauts Story Musgrave and Jeffrey Hoffman practice removing the Wide-Field Planetary Camera (WF/PC) unit in the Weightless Environment Training Facility (WETF) water tank facility at JSC. In the bottom photo, Astronaut Thomas Akers rehearses for the HST repair mission. Note that in both photos other divers help to make the movement in the viscous fluid more closely resemble the vacuum of space.

Some of the goals of the *Integrated EVA/RMS Virtual Reality Simulation Trainer* included:

- “Evaluate the capabilities and limitations of VR technology to support integrated EVA training in a useful way.
- Provide overall situational awareness of on-orbit EVA operations as they related to water tank training configurations.

- Develop a command protocol for communications between the EVA astronaut and the RMS operator.” [Homan]

Each EVA offered a different scenario in which astronauts would interact. The first HST repair mission (STS-61) involved two EVA astronauts working together with an IVA astronaut. For this mission, eight 2-hour sessions were conducted in the VR simulator. During these eight sessions, a command protocol between the EVA and RMS operator was developed and the tasks of removing the High Speed Photometer (HSP), moving it to a temporary holding device, removing the COSTAR (Corrective Optics Space Telescope Axial Replacement) from a replaceable unit carrier and placing it in the HST, and then storing the HSP into the replaceable unit carrier were practiced.

The objects modeled for the HST training application included all the objects with which the astronauts would interact. The design team constructed a computer model of the HST, the payload bay and fixtures used for transporting replacement systems into orbit, and the components that would be replaced or serviced during the repair and maintenance mission. The rest of the Space Shuttle components (that would not be seen during this task) were not modeled. Objects were created using the engineering drawings of the real-life counterparts, for more accurate representation.

Viewing the world was also implemented in a way to mimic real-world viewing conditions. For tasks such as the wide-field planetary camera replacement task, the extravehicular astronaut would be physically immersed in the VR system, while the astronaut performing the intravehicular activity would sit in front of a computer monitor that provided the portal view to the external shuttle bay. The external astronaut would interact with the environment with their hands, while the internal astronaut would manipulate the shuttle’s robot arm (called the Remote Manipulator System or RMS) using joystick controls similar to the controls on the shuttle. The IVA astronaut could view the world either via a window looking into the payload bay, or on a pair of monitors that show the world through the view of cameras mounted throughout the shuttle. Four cameras are located in the payload bay (one in each corner), two are mounted on the robot arm, and others are placed in various locations in the shuttle (or are simply handheld).



FIGURE 7-19

This computer rendering resembles the graphics presented by the virtual reality system. It depicts the Hubble Space Telescope attached to the Space Shuttle as two astronauts perform repairs.

The physics of the virtual world were programmed to closely match the low-gravity environment that is experienced by the astronauts. Virtual objects were constrained to move within the limitations of the real objects. For example, when extracting long objects from the telescope, the astronaut would have to continue to guide and pull the object until completely removed.

EVA astronauts maneuver in one of two primary ways. One way was to be attached to the end of the robot arm and be moved about by an astronaut controlling the arm from within the cabin, much like a worker operating atop a “cherry picker.” The other method was to be a “free floater,” crawling along the structure using handrails. Handrails are placed frequently on the external structures of objects about which an astronaut may need to maneuver.

Movement in the virtual world is carried out in just the same way. Astronauts attached to the RMS move as it is maneuvered by an IVA astronaut, while free floaters pull themselves through the world by “grabbing” handles and pulling. The grabbing is done by making fist gestures that are sensed by the data gloves worn by the astronauts. Objects are maneuvered by manipulating tracked physical objects. A tracked prop is used for low-mass objects, and



FIGURE 7-20

Tricia Mack (EVA trainer) and Jeff Hoblit (VR simulator programmer) demonstrate the VR training system as used for STS-88.

the Kinesthetic Application of Mechanical Force Reflection (KAMFR) system (described below) is used for large objects.

Extravehicular activities are always done with at least two astronauts participating. Frequently (and as was the case for the first HST repair mission), one extravehicular astronaut is attached to the RMS, while the other floats freely. The RMS astronaut typically moves large objects into place, while the free floater performs tasks such as getting equipment out of storage. In STS-61, astronauts Jeff Hoffman, Thomas Ackers, Kathryn Thornton, and Story Musgrave were teamed in pairs, with two performing the EVA each day. Mission specialist Claude Nicollier was the primary operator of the RMS during the mission.

**FIGURE 7-21**

One of the EVA astronauts rides at the end of the RMS while the other moves by grasping handles on the shuttle and devices.

To prepare for STS-61, EVA astronauts took turns in the HMD to experience segments of the HST repair. At the same time, Nicollier would operate the virtual RMS, and the other astronauts could watch and comment on the activities taking place in the virtual world. Movement of the virtual robot arm had the same constraints as the real thing. By putting realistic constraints on the virtual world, the trainees get a chance to not only see what the operation will be like, but also to get a chance to experience a realistic response to their actions.

Realistic avatars represent each astronaut's position in the virtual world. Working together requires that each participant knows where the other participants are and what they are doing. Astronauts also need to see their own body, so they can interact with the virtual objects. However, each astronaut's body is rendered differently from the user's perspective versus that of their crewmates. Astronauts see their own arms, hands, fingers and the inside of their EVA suit helmet. Others see the entire body of the astronaut, but the hands and fingers are not individually articulated. To reduce polygons, the entire hand was modeled as a single "mitten."

Audio in the virtual world was primarily voice communication between astronauts. Because all the sounds that an astronaut performing an EVA will hear will be those

**FIGURE 7-22**

Inside the shuttle, mission specialist Claude Nicollier operates the RMS controls while viewing the action through four portals and two CCTV monitors.

coming over their communications link, there is no need to add any sounds other than normal voice communication to the experience.

VR System

The *EVA/RMS VR simulator* provides the hardware interface to a virtual world that mimics the physical behaviors of objects and machinery in space. This world includes "a functionally correct dynamic simulation of the Shuttle Remote Manipulator System with translational and rotational hand controllers, graphical representation of the A8 control panel, and computer graphic renderings of the objects important to the scenario. The simulation also provides access to all closed circuit television (CCTV) views available from all payload bay and RMS cameras and out-the-window views" [Homan and Gott].

An HMD visual display was used for the physically immersed participants. A single Virtual Research VR-2 Flight Helmet was used in the early research. More recently, the JSC VR lab upgraded to using a pair of n-Vision high-resolution (1280 × 1024), CRT-based, head-mounted displays. These HMDs provide wide (120-degree) horizontal field of view with 50% stereo overlap. The vertical FOV is 80 degrees. After the success of the first mission, a second HMD was added to the system so both of the EVA astronauts can be immersed together in the virtual space.



FIGURE 7-23

Astronaut Jim Newman prepares for his EVA to begin constructing the Unity Space Station on STS-88. He wears a high-resolution HMD to view the world, and two data gloves to manipulate the world. Newman interacts with astronaut Jerry Ross (seated behind him) in the same virtual world.

Along with the HMD, physically immersed astronauts wear two data gloves for interacting with the world, while sitting in a chair. The IVA astronaut uses a standard computer screen for monitoring the external actions, and physical and virtual controllers to manipulate the robot arm.

Position tracking is done with an Ascension Flock-of-Birds™ electromagnetic tracking system. Ten sensors were available for tracking the head, both hands, torso, and a handheld prop for each of two immersed participants. The handheld prop was used as an interface to represent lightweight objects being manipulated by the astronauts.

A large, Silicon Graphics Reality Monster™ graphics supercomputer provided the rendering power needed for the more complex rendering tasks required by the simulations. The Reality Monster provided several visual outputs. Typically, four outputs were used to drive the two head-mounted displays (two for each HMD). Another visual output shows an out the window view from within the shuttle

(looking into the payload bay). Two other monitors mimic the camera monitors on the shuttle.

The prototype work done for STS-61 used an older (and therefore less powerful) Silicon Graphics workstation, with only a single HMD (used by the EVA astronaut) and Polhemus position sensors.

For communication between astronauts, much simpler technology was used. For activities taking place within the VR facility, an amplification system with headsets was used. Each astronaut in the crew, plus trainers, wore their own headset. The number of participants in the conversation is only limited by the number of headsets. For collaborations with participants in different locations, a telephone conferencing system was used. It offered the best performance for the cost and effort.

For the purpose of debriefing evaluations of trial runs in the VR simulator, the system records all the motion and other input data, as well as the voice communications between astronauts. Playback of past VR experiences can be controlled interactively to view a particular segment of time, or change a camera view.

A very important element in learning how to maneuver oneself and objects in microgravity is learning the effects of inertia from an experiential point of view. One can learn the formulae for how objects behave in space, but developing an instinct for how to work in such an environment is best done through experience. Of course, much of this experience comes from haptic interactions.

About a year after the Hubble Space Telescope repair flight (STS-61), a *mass handling simulation* system was developed at the JSC. This system consisted of a box hung on cables that are computer controlled to apply forces to the box. This system, named Kinesthetic Application of Mechanical Force Reflection (KAMFR), was built using preexisting “Charlotte” robots at JSC. The Charlotte robots were designed for use to control operations in a space vehicle or space station during crew sleep periods. The JSC VR team adapted it for use as a haptic I/O device. The “tendon-driven” KAMFR system was used to emulate the physics of objects with large mass (greater than 180 kg). As the astronaut pushes on an object haptically mimicked by the *mass handling* system, the motion dynamics of the object are calculated,

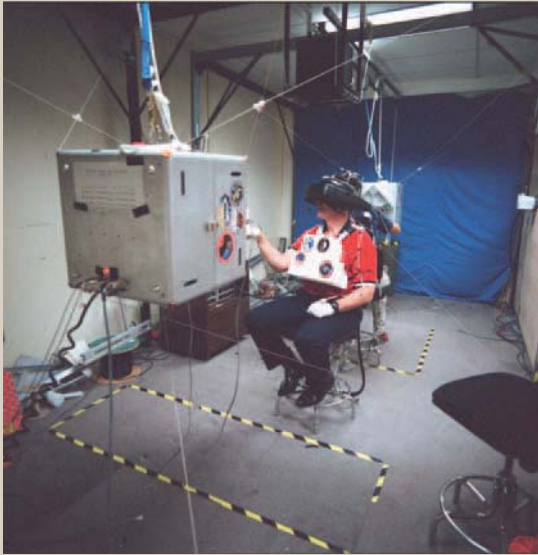


FIGURE 7-24

The dual KAMFR haptic display systems can be seen in the foreground, and far background, as astronauts Jim Newman and Jerry Ross train for STS-88.

and the robots pull on the appropriate cables to mimic the resultant forces.

Because EVA astronauta are wearing an HMD, they don't see the box on cables. What they see is whatever object the box is currently mimicking.

One way in which the KAMFR system is used is to train how to manually capture objects not stable enough for the RMS to grasp. For example, in STS-87 (November 1997), two EVA astronauts captured the *Spartan* satellite by hand after it had been set tumbling when the *RMS* bumped it. Prior to the mission, these astronauts had trained for this contingency with the KAMFR system. The system allowed them to practice handling the 3000lb payload as part of their training.

The *mass handling simulation* was first used for preparation of STS-63. It was later used for the second HST repair mission (STS-82 in February 1996), and was also used for many other shuttle flights involving EVA operations. The evaluation of the EVA crew of STS-63 was that the KAMFR

system "successfully replicated the zero-g mass characteristics of the 2700lb *Spartan* payload manually handled during the EVA Development Flight Test" [Homan and Gott]. Later, in November of 1996 (STS-80), two KAMFR systems were integrated to test the ability of the system to allow two astronauts to manipulate a single large object.

System Development

Programming for the *EVA/RMS VR simulator* was done in-house. The software core was the *OpenGL* graphics library and the C programming language. In addition to the main training applications, the JSC team developed supporting tools for creating the models, environment layout, behaviors, and feedback.

However, the simulation system is never static. When crews come to the facility to train for an upcoming mission, they use the most recently developed version of the software. The software is continually updated to include features necessary for future missions, and also to accommodate changes requested by the astronauts. Of course, this means that this is not a stand-alone system—some of the developers must always be on-hand when the system is being used to operate it.

The JSC VR simulation group is led by David Homan. Other programming and support of the VR facility are provided by contractors from LinCom Corporation. These people are Brad Bell (who has worked in the lab since its inception and works on the graphics software), Jeff Hoblit and Ken Huffman (who integrate all the dynamics simulations), and Erin Orgeron (who builds the models from engineering drawings).

While they had been experimenting with virtual reality since 1988, they began development on the *Integrated EVA/RMS Virtual Reality Simulation* about 1 year before the launch of STS-61.

User Perspective

The test of any virtual reality training application is whether it is found to be useful both through formal analysis and whether the user's themselves find the experience worthwhile. In the case of the training provided by the Integrated EVA/RMS Virtual Reality Simulation Facility, astronauts have found the facility useful for a particular subset of extravehicular training.

Astronaut Dr. James Newman found the VR facility to provide complementary training with the Neutral Buoyancy Lab (NBL). The NBL is a new, larger water tank for practicing space walks under water. But despite the increased size of the new facility, the tank is not large enough to hold complete units of the new space station, thus limiting some aspects of training.

Parts of the EVA experience that are best learned in the NBL include learning how to “work with the suit” and learning to perform precise manual tasks such as connecting cables between adjoining modules of the station.

Because the space suits are airtight, the occupant is limited in how they can maneuver the suit. For example, bending an arm might force air out of that sleeve and into another section of the suit. Thus if the astronaut wishes to constrict both arms, they may not be able to do so because there won't be any place for the displaced air to move.

Virtual reality technology has not yet advanced far enough for simulating full use of all the fingers on the hands, complete with haptic feedback. So, when trying to learn to accomplish procedures that require sufficiently learned dexterous skills with the hands, there is no substitute for wearing the actual glove in an environment where the objects one is working with might float away. On the other hand, the viscosity of the water is different than the vacuum of space, but it is still the closest one can get to simulate this task while not in space.

The virtual reality experience has been found by Newman and the other astronauts to surpass the water tank for activities such as allowing the EVA and IVA astronauts to work together, allowing the EVA astronauts to practice navigating from location to location, and practicing operation of the SAFER unit (described below).

One of the limitations in working in the water tank is that the EVA astronauts and the RMS operator cannot practice their operations together. RMS practice in the water tank involves the use of a dummy astronaut attached to the end of the RMS. The RMS operator can then practice how to maneuver the arm to put the dummy into a useful position, but the dummy cannot give feedback as to how satisfactory the position is for performing the necessary chore. In the virtual reality experience, an EVA astronaut can ride

along as the IVA astronaut moves the arm, and they can work together to find a good position and, more importantly, how best to communicate with one another during the procedure.

The EVA astronauts can also benefit from becoming familiar with how they will move from place to place while they are free floating. In space, there is no fixed horizon about which all movements can be referenced. Thus while free floating, astronauts need to be more deliberate in how they move about. By spending hours in the VR simulator, they can become familiar with the layout of the units there are working on and how best to get from one place to another.

Where VR really shines is in training for use of the SAFER (Simplified Aid For EVA Rescue) unit. SAFER is a backpack device with compressed air that allows the astronaut to maneuver in space should they become untethered and out of reach of handholds. If an accident occurs that results in an astronaut becoming detached from the shuttle or space station, they can fire the compressed air to correct their position and fly back. However, the limited amount of air in the unit may not permit the astronaut to maneuver with the unit and then return to the craft. They must know how to operate SAFER before their actual trek into space. The VR training facility makes this possible. The virtual reality experience also allows the astronauts to do sanity checks on operations that were practiced in the water tank to discover if any of the planned maneuvers are unrealistic.

Another test of how well a VR simulation works as a means of training is whether the trainees lose themselves in the experience and become immersed. Newman reports that the weight of the HMD does serve to remind him that he was not in the weightlessness of space, and he found that even with counterbalances added to the HMDs, it was good to take a break after about an hour of use: “It's hard to lose yourself in the virtual reality experience when there is this thing crushing your skull.”

On the other hand, Newman did find that the SAFER training in particular provided an experience in which all his focus was on the experience:

When you get closer to flight, and you know this is like the best training you're gonna get, and if you die in the sim, that doesn't bode well, so the

adrenaline ... really gets going. And I would say that's the closest I've come to getting lost in the simulation, with the prospect of my own demise, and what I do next makes it a real difference.

If I were to go fly again on flights this year, then I wouldn't feel the need to go into the tank near as often as we would have wanted before having done this last flight. Because there is a huge amount you learn in the tank, but once you've been outside three, or more times. Now I would be very comfortable doing some VR lab sims, a couple of NBL runs and then [I'd] be ready to go.

Related Work

At the same time the JSC VR lab was coming to fruition under the guidance of David Homan, Bowen Loftin at the University of Houston was working with the Johnson Space Center ISD Software Technology Branch, the Mission Operations Directorate (MOD) space flight training division, and the MOD flight director's office to study how virtual reality might help flight controllers understand the shuttle missions from the perspective of an astronaut. In this study, a VR system was produced independently of the JSC VR lab for use by ground-based personnel.³

Normally, due to the limited availability of training resources and personnel, few flight controllers have the opportunity to undergo mission training. Resources for the important hands-on training are heavily scheduled and expensive to operate. Thus, usually only the astronauts had access to the training facilities and simulations. The ground-based flight controllers and support team have much less (virtually no) opportunity for such training.

The goals of the *Flight Team VR Training System* were somewhat different from those of the Astronaut trainer [Loftin and Kenney]:

- Know the correct sequence of EVA activities
- Have an awareness of the position and orientation of the HST, fixtures, components and astronauts

- Be able to identify hardware components
- Perform specific tasks associated with the repair/maintenance of individual components such as bolts to be removed in sequence

Whereas astronauts were using VR (and traditional mission-readiness techniques) to prepare for the HST repair, Loftin's goal was to provide an opportunity for key members of the ground-based support team to interact with and learn about the operability of the various components on the HST. These flight controllers, engineers, and technicians were also trained in the related maintenance components in the payload bay.

For STS-61, several steps were involved in repairing the HST, so the VR practice task was divided into six (EVA) scenarios. The level of detail implemented for each scenario was determined by the intricacy of the procedure. The scenarios included [Loftin and Kenney]:

- Solar Array change-outs
- Rate Sensor Unit (RSU) change-out
- Corrective Optics Space Telescope Axial Replacement (COSTAR)
- Wide Field/Planetary Camera (WF/PC II) change-out
- Solar Array Drive Adapter Electronics (SADE) replacement
- Magnetic Sensing System (MSS)-Magnetometer installation over original magnetometers

The 105 flight-team members who used the system were very highly motivated. After the training was completed, the developers conducted a survey (38 were returned). The survey intended to study three facets of the program. First, they wished to study the effectiveness of this training in enhancing the flight team's performance during the HST mission. They also sought to evaluate the training potential of VR technology, as well as to assess some of the human factors issues and user-to-environment interface methodologies afforded by the training system.

Overall, the participants liked the training and thought it was enjoyable. The positive comments greatly outnumbered the negative. They thought the training was efficient and that they could compress the time required for simulating

³*Project ScienceSpace* (Chapter 6) describes additional work performed at the University of Houston.



FIGURE 7-25

Flight controller Paige Lucas uses VR system to have an opportunity to experience an abbreviated HST repair EVA.

a multi-hour EVA to less than an hour. Some of the negative comments referred to the slow response from hand movements and the resulting visual affect. Users disliked the weight of the HMD, and reported several problems with the data glove used for interfacing with the virtual world [Kenney and Saito].

Many of the problems with the data glove resulted from hand calibration. Calibrating a glove for a particular individual's hand is difficult and requires some setup time prior to users being able to enter the experience. If the system is not properly calibrated to each user's hand, the system can have difficulty recognizing when participants are attempting to trigger an event with a preprogrammed posture. For example, when required to make a fist, many people were unsuccessful or found that they had to squeeze their hand closed as tightly as possible to activate the posture recognition. This led to hand fatigue. A quick solution that Loftin's group discovered was to have a premade suite of calibrations for various hand sizes. By choosing a hand-size that matched the user's hand,

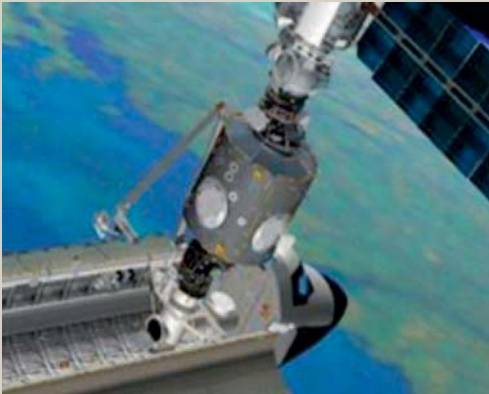
a decent calibration setting could quickly be attained. When the glove was correctly calibrated, it was very easy to activate the fist operation.

The development on the *Flight Team VR Training System* to be used for the HST repair mission began in May 1993. Three of six planned scenarios were done by September 1993, at which point the training with the system began. By October 1993, the rest of the scenarios were completed. Throughout the duration of the training, developers continued making enhancements and improvements until the mission was deployed in December, 1993. In all, the development of the initial system took 6½ months and utilized the talents of six full-time developers and three university interns. Roughly a third of the time was spent doing data acquisition from drawings, and video of details of the specific procedures.

Loftin's team achieved their goal of giving the ground-based flight team a better idea of the situations the astronauts would contend with during the mission. In the process they trained over 120 people in the VR simulator. The large scope of this project provided a unique opportunity to gather data on the usefulness of VR as a training tool. The main finding was that the flight team found the VR training experiences to enhance their performance during the flight of the mission. Based on the success of this particular training application, the *CyberEdge Journal* awarded the JSC VR team the 1993 CJ award for virtual reality applications.

As mentioned, the *Integrated EVA/RMS Virtual Reality Simulation* was a useful tool in the preparation of shuttle missions.

More importantly, the success of the *EVA/RMS VR simulator* by the JSC researchers can be proclaimed by the fact that the system has been used on subsequent EVA missions since it was first used in 1993. For example, STS-88, flown in December 1998, involved three EVAs to construct the first components of the international space station. In addition to bringing the large components together in space, there were a lot of activities such as attaching connectors and connecting cables. In preparation for this mission, over 150 hours were spent in the *EVA/RMS VR simulator*.

**FIGURE 7-26**

During STS-88, astronauts James H. Newman (left) and Jerry L. Ross work to connect the Russian Zarya and United States' Unity modules together.

Once the space station is operational, it too will have an RMS arm, and will be used to conduct EVA operations using the space station as the base. The VR simulation program will then be useful to simulate these missions as well.

Conclusion

It is worth reiterating that there are activities that virtual reality astronaut training provides a better method of experiencing and therefore better mission preparation than the traditional methods of experiential training. For example, VR provides the best means of learning to fly the SAFER “backpack” units that allow astronauts to fly themselves back to the shuttle if their tether ever becomes severed.

Also, the VR simulator is the only way that the entire crew can go through entire scenarios together. The water tank and other low-gravity simulators generally only allow either the EVA astronauts or the IVA astronaut to practice their tasks individually, not as a team.

When asked whether he sees VR playing an increased role in astronaut training, Newman stated:

I see an increasing role, and I do believe in truth, that particularly when you have experienced people—I don't need to go into the NBL near as much now, in my opinion, as I would have wanted to prior to this flight. I can do a lot of it in the VR-lab, and it's a lot cheaper to do it that way.

CASE STUDY 7.5: INFANTRY TRAINING (NPS/ARL)

Computers are increasingly being used to mediate warfare practice exercises. To create simulations of large-scale operations requires the use of a large number of computers at a training center. The potential benefits of computer mediated warfare training are the ability to bring together

units physically located in disparate regions of the globe, and to do exercises using a far less dangerous, and environmentally disruptive methodology. However, such training is not only worthless, but potentially detrimental if it is not designed to eliminate negative training brought on



FIGURE 7-27

A soldier moves through a virtual world by walking on the Sarcos Treadport I/O device at the AUSA 1995 conference. (Photo by Michael Zyda)

by interfaces that allow the trainee easier interactions, or more information than available in the real world scenario.

At the Naval Postgraduate School (NPS) in Monterey, California, a large virtual environment simulation system was developed using the, then current, DIS protocol for collaborative communications, *NPSNET* [Zyda et al. 1993]. By freely distributing *NPSNET* on the Internet, NPS was able to test the system with users at numerous locations, operating on different systems with a variety of input and output devices. Over the course of development, many large-scale exercises were demonstrated.

A limitation of the early system was that it was heavily geared toward operations involving ground and air vehicles, but not dismounted infantry (DI). Two reasons for this are that it is much easier to create interfaces that allow participants to control vehicles in ways similar to the real world, and that the creation of Semi-Automated Forces (SAFs) used to populate the world with a greater number of opposing forces were easier to implement for vehicle-based operations. In short, a DI is harder to emulate.

Specifically, it is difficult to achieve a computer-mediated interface that effectively simulates the conditions a soldier would face in the real world. For instance, vehicles can be operated with a set of controls and viewing parameters similar to the actual vehicle. Also, constraints such as fuel

consumption and maximum speeds up and down hills can be realistically imposed by the computer simulation. On the other hand, it is more difficult for a computer to know the level of exhaustion a dismounted soldier would experience and how fast they would be able to move at any given time. Interface devices that provide natural interaction for traveling through a virtual world as a dismounted infantryman (such as physical locomotion with feedback based on the terrain) have been few and far between, and those have been mostly unsatisfactory.

The NPSNET Research Group has investigated several types of input/output devices using their *NPSNET* software connected to hardware provided by collaborators such as Sarcos Research Corporation. A “Soldier Station” described by Shirley Pratt and colleagues, provides a joystick and button interface that is useful for people in the virtual world with enemy forces [Pratt et al.], but because it can’t truly simulate a soldier’s exhaustion, it doesn’t succeed as a method of simulating how the differing amount of physical exertion required between two options will affect choices made by the soldier. For example, a soldier may be able to choose two paths to get from one point to another. The path with less chance of encountering an enemy might require a physically tiring climb and descent of a hill, while an alternate route may be physically easier but have a slightly higher chance of enemy encounter. If the only effort required to ascend and descend the hill is to move a joystick in a particular direction, the soldier will obviously choose the more cautious route. However, if a soldier tired from a long march is faced with this decision, knowing considerable effort will be needed to climb the hill, the (more dangerous) level route may be chosen.

The Army Research Laboratory (ARL) is interested in finding a more realistic interface for DI interactions with simulated warfare environments. The ARL sought to replace the traditional ICSAF (Individual Combatant Semi-Automated Force)⁴ with hardware systems designed to fatigue a participant as they traverse terrain “on foot.” These systems were referred to as Individual Portals, or I-Ports. As part of their research, the ARL funded the NPS to demonstrate some scenarios that make use of new I-Port systems. Three demonstrations were presented to Army personnel at the

⁴Basically “Wizard of Oz”-style agents.



FIGURE 7-28

At Fort Benning, a soldier (left) moves through a virtual world using the Sarcos UniPort for the February 1994 demonstration of NPSNET-IV at the AUSA conference and (right) a soldier moving through an NPSNET-IV virtual world on the Sarcos TreadPort. Left image courtesy of Michael Zyda and the right image courtesy of Sarcos

annual Association of the United States Army (AUSA) Conferences in Washington, D.C. from 1994 to 1996.

Application Description

In the AUSA 1995 demonstration scenario, a squad was called upon to eliminate a sniper firing from a building. The plan was for the team to get into the building, up the stairs, and shoot the sniper. Five participants were involved in this scenario: the enemy sniper, the squad leader, and a three-member M60 machine gun team. The soldiers controlled the simulation using a variety of interface methods. Two of the

machine gun soldiers were traditional ICSAFs (controlled by someone behind the scenes), and the other three participants were stationed on different I-Port input devices: the UniPort, the TreadPort, and the StickPort [Pratt et al. 1995]. The sniper operated a UniPort, a unicycle like device that can provide a variety of resistances to the pedaling operation. The squad leader used a TreadPort, a unidirectional treadmill device, and the third member of the “fire team” used a StickPort, a joystick and button input interface.

After receiving fire from the sniper, the three members of the team provided protection by firing M60 machine guns

as cover fire. The squad leader would then stand up and use a hand signal to communicate to the others that they should “Follow me.” The two semi-automated (ICSAF) participants would then run to the building, followed by the StickPort participant, and finally the squad leader using the TreadPort. Once at the building, the fire team would follow the leader up the stairs. After navigating through rooms, and around furniture acting as obstacles, the team could then shoot the sniper. After this task was accomplished, one of the soldiers would wave the “All Clear” signal from the window where the sniper had been.

In NPSNET, if a dismounted soldier were virtually killed, they would see a red screen, and then view the world from a prone (lying down) position. Soldiers would return from the dead after 30 seconds.

In many dismounted military maneuvers, communication is done using silent hand gestures. The generation of these signals can be done with two different techniques: button presses that cause the avatar to move through a preset movement pattern that depicts the gesture, or by letting participants gesture naturally, with their own body. The latter method requires that enough of the participant’s body be tracked to articulate their avatar in the appropriate way.

The conference audience for this particular demonstration was high-level military officials who needed to see some real benefit from using the new technology over current (proven) methods. In addition to the conference venue, the AUSA 1995 demonstration also included a network connection to Ft. Benning in Georgia, which is a location where such operations would be more likely to take place. In the demonstration described here, the participant acting as the sniper was located in Ft. Benning, and used the UniPort input device.

The scenario used for this presentation was a rather short operation based on the time constraints of quickly demonstrating the possibilities of the technology. Each of the I-Port devices supports only a single individual in the environment, so several devices would be necessary for larger operations. Plus, in actual warfare exercises, the participating soldiers would need to be “suited up” during the entire (generally more lengthy) mission.

VR System

The Naval Postgraduate School received funding to demonstrate the fourth major version of their software suite, *NPSNET-IV*, integrated with new input/output technologies and designed as a training tool for military operations. The hardware and the software involved were both key parts of what was being demonstrated. In the 1994 AUSA demo of *NPSNET-IV*, a SensorSuit was used to track the participant using mechanical means, and an HMD was used to display the visuals. For the 1995 demonstration, the UniPort device was integrated into the demonstration. In addition to this new device, other hardware changes from the 1994 demonstration included replacing the SensorSuit with a Polhemus FaskTrak™ electromagnetic tracking system, and using projectors rather than HMDs to display the visuals. New features of the *NPSNET* software were also highlighted in 1995.

VR DISCOVERY

The switch from HMD to a projection-based display was made because when soldiers used the HMD system they would routinely bump the HMD with their rifles.

The SensorSuit, which had previously been used for testing computer-mediated DI operations, was found to interfere with some of the postures necessary in conducting military operations. The SensorSuit used mechanical linkages to measure the joint angles of the participant. These angles were then used to reconstruct their avatar in the virtual world in the appropriate posture. A Polhemus FaskTrak™ electromagnetic tracking system with four receivers provided a reasonable alternative to the SensorSuit. Another disadvantage of the SensorSuit was that it took more than an hour to calibrate the SensorSuit for each user.

The Polhemus system did not provide as much detail over the entire body of the participant, but did provide 6-DOF position of four points on the user’s body that could be used to approximate their current posture. The four receivers were used to track the wrists and head of the participant, plus their M60 rifle. When an HMD is used to display

the visuals, the head sensor was mounted directly on the display. However, for the projection display, it is less important to know the orientation of the head than the location, so a point at the base of the neck was chosen for tracking the head's location as well as the torso of the participant. On the rifle, the tracking sensor was mounted on the stock. The developers would have preferred to mount the sensor along the sight line, but were prevented by the design of the rifle mock-up.

When the participant was holding the rifle, the *NPSNET* software was programmed to automatically snap the hand of the avatar to the rifle avatar. By doing this, participants in the world would not see the hand and rifle move slightly independently as a result of jitter caused to the electromagnetic tracking system by the TreadPort and UniPort devices.

The TreadPort and UniPort motion devices were both used to provide a means of participant input generated by leg motion. Leg motion input serves two purposes. It more closely resembles the method of locomotion used in the real world, and it causes the participant to physically exert themselves to travel over the terrain.

The TreadPort allows the user to walk on a treadmill that can sense the distance they were effectively traversing. The UniPort allows the user to sit on a device with a set of bicycle-like pedals. Rotating the pedals provides the mechanism for traveling through the virtual space, and can also provide feedback in the form of resistance to convey the difficulty of traversing particular terrain, thus requiring more effort to go up a hill than to walk across a flat region. Turning in the virtual world on the UniPort was (awkwardly) done by the user rotating their knees to the direction they would like to travel. On the TreadPort, the system would sense when user attempted to walk in another direction and rotate the scene in the visual display, and the attached robot arm would return them to the center of the treadmill.

A BG Systems Flybox™ input device provided the controls for manipulating the ICSAF characters in the scenario.

Communication between sites collaborating in the same scenario was done using the Military's Distributed Interactive Simulation (DIS) protocol. This protocol was developed to handle communication between a large

number of computers across wide distances for the purpose of creating a collaborative virtual world to enact military operations. Multicasting of the DIS information was done using the standard Mbone network protocols [Macedonia et al.]. The DIS protocol was a descendent of the SIMNET protocol, and has now been superseded by the HLA (High Level Architecture) specification.

A separate NPSNET-3D sound server [Storms et al.] provided audio feedback. The sound server software renders the sounds that correspond with actions in the world such as vehicle engines and gunshots or explosions. Sounds are filtered based on the distance and speed of the object generating the sound from the listener. The NPSNET-3D sound server uses MIDI devices (sound samplers and effects units) to control the sounds.

Vocal communication is handled separately from the sounds generated by the virtual world. In the 1995 sniper scenario demonstration, communication was done via hand gestures, so no extra effort was required. When communication is required (such as the 1994 AUSA demo) actual radios that duplicate the real-world equivalent are used.

The *NPSNET-IV* collaborative virtual environment software system is built on the Silicon Graphics *Performer*™ library for the primary scene generation and thus runs on (and only on) the full range of Silicon Graphics workstations. Human models for the 1994 and 1995 demonstrations were created using the *JACK-ML* motion library (a trimmed down, real-time version of the *JACK* system from the University of Pennsylvania). Later, Boston Dynamics' DI-Guy was used by *NPSNET-IV* for the human models.

Since *NPSNET* was originally designed with vehicular movement in mind, some reprogramming was required to correctly model DI movement through the world. In particular, *NPSNET* provides a terrain-following constraint for ground-based vehicles and dismounted infantry. However, because it was assumed that vehicles would not be entering buildings, the terrain-following would place objects on top of any objects with which they are copositioned. Thus, tanks and people would always jump to the top of any building they attempted to enter. To allow individuals to properly traverse buildings (including multistory buildings), the terrain-following constraint was modified to

position the user on the nearest surface below their head rather than the highest surface at a given latitude and longitude. This modification also allowed participants to correctly move up and down stairs inside of a building.

The AUSA demonstrations were developed in conjunction with SARCOS Research Corporation (SRC). The overall system was a collaboration between NPS, SRC, the Army Research Lab, the Simulation, Training, and Instrumentation Command (STRICOM), Hughes Research, and the Dismounted Battle-space Battle Laboratory. The TreadPort and UniPort devices were created by SARCOS, and the Naval Postgraduate School integrated the systems to create the demonstration system. The 82nd airborne provided the participants who were engaged in the virtual battle.

At the Naval Postgraduate School, Professor Michael Zyda led the *NPSNET* Research Group. The main group of programmers responsible for the development of the *NPSNET* software system includes Shirley Pratt, Paul Barham, Randal Barker, Scott McMillan, and David Pratt.

Conclusion

The AUSA 1995 DI scenario provided a good demonstration of how dismounted infantry could be integrated into computer-mediated warfare exercises. However, neither the UniPort nor TreadPort devices have been widely deployed as part of the military's training program. This may be due to the cost and difficulty of outfitting each individual DI with their own system.

Although no formal studies have been performed comparing the physical exertion devices (the TreadPort and UniPort) with simpler joystick devices, there has been some comparison of traditional route training methods with virtual reality. In particular, Witmer and colleagues used traversing a particular route through a building to compare traditional visual route training methods with VR and experience in the actual space [Witmer et al.]. The results indicate that practice in the actual location results in the best route learning, while VR outperformed the use of photographic and other presentations of the route (See the discussion on the *Shadwell* firefighter training also in this chapter).

Besides the sniper simulation described here, systems with DI input devices connected to a collaborative virtual



FIGURE 7-29

An omnidirectional treadmill allows participants to walk in any arbitrary direction, and allows maneuvers such as sidestepping.

environment could be used (and have been demonstrated) for a variety of other applications such as hostage rescues, battle scenarios, police and fire training, etc.

The 1996 *NPSNET-IV* demonstration at the AUSA conference was used to introduce yet another ground-breaking I/O device—the Omni-Directional Treadmill (ODT), developed by Virtual Space Devices Inc. The ODT was created to accomplish the same result as the UniPort and TreadPort devices—to duplicate the physical exertion of traversing an environment on foot. However the ODT was designed to overcome some of the problems associated with those devices. In particular their inability to allow maneuvers such as sidestepping.

NPS and Virtual Space Devices collaborated to investigate how well an ODT could emulate real-world motions by the participant [Darken et al.]. They found that after practice many users could operate with the device, but with many restrictions. Because the device always moves to keep the user in the center of the platform, there are many circumstances that can cause the participant to stumble. Adaptation to the device generally results in users learning what circumstances are likely to cause them to lose their balance, and thus they learn to avoid these cases. Unfortunately, some of these circumstances are maneuvers that they would normally use in the real world—such as moving laterally. Also, activities such as crawling or

squatting are not possible because a tether on the user will sense this and interpret it as the user falling over, and thus will kill the motors to the ODT.

As is frequently the case, a lack of sustained funding is sometimes the biggest problem encountered during the development of large VR systems. As Zyda describes the fate of *NPSNET*:

The major limitation of the 1994–1996 AUSA demonstrations was that the work never lead to anything fundamentally reusable. Instead of

providing continuous funding of the work during the 3-year demo cycle, money was provided only to slap together short-lived demonstrations. That money would be provided maybe 60 days out from the demo. The short life for the funding did not allow the full development of articulated humans in the NPSNET-IV platform and did not allow for their full testing and support. The final AUSA 1996 demo was the end of development on NPSNET-IV due to this lack of funding constancy and due to demonstration-fatigue of the NPSNET-IV team.